

Shreya Pramanik

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Summary

Statistics M.S. graduate from the University of Minnesota with hands-on experience deploying XGBoost, LSTM, and Prophet ensembles for demand forecasting at HSBC, achieving 80–86% accuracy across 5+ APAC contact centers. Built end-to-end ML pipelines from feature engineering through model evaluation across deep learning domains, including multi-modal EEG classification and hospital financial stress-testing, achieving up to 49% improvement over single-source baselines.

Technical Skills

Languages & Frameworks: Python, R, SQL (MySQL, Google BigQuery), SAS, MATLAB, Excel, Power BI (DAX, Power Query), Tableau, PyTorch, TensorFlow, scikit-learn

Cloud & DevOps Tools: AWS, Azure, Databricks, Docker, Snowflake, Git, Airflow, n8n

Data Tools: ETL Pipelines, Salesforce, CRM Analytics

Experience

HSBC Electronic Data Processing Pvt. Ltd

July 2022 – July 2023

Data Analyst, Wealth and Personal Banking

Bengaluru, India

- **Queried** and extracted large-scale customer contact datasets via **SQL** across 5+ APAC contact centers; built Excel-based reporting tools using **pivot tables**, **lookup formulas**, and automated **KPI summaries** to clean, structure, and surface trends.
- Developed and deployed an ensemble of short-term demand forecasting models using **XGBoost**, **LSTM**, and **Prophet** across 5+ APAC contact centers, achieving **80–86%** accuracy, informing real-time staffing decisions for hundreds of daily customer interactions.
- Led production-ready modeling pipeline on Malaysia customer-contact data to detect demand surges during digital-banking rollouts, informing **workforce planning decisions** to business stakeholders.

School of Statistics, University of Minnesota

Sep 2023 – Jan 2026

Graduate Teaching Assistant

Minneapolis, MN

- Led coding labs for regression and simulation courses; mentored 100+ grad students on statistical programming in R.

Carlson School of Management, University of Minnesota

Feb 2026 – May 2026

Graduate Research Assistant

Minneapolis, MN

- Cleaned 654 survey responses (flagged ~5% low-quality) and coded 16 interview transcripts (**Cohen's $\kappa = 0.78$**) for a longitudinal reduced work-week policy study.

Projects

E-Commerce Analytics Platform | *MySQL, Python, Power BI, DAX* ➡

Jun 2026 – Jul 2026

- Built an end-to-end **ETL pipeline** using Python (pandas, SQLAlchemy) to ingest **100K+ real transactions** across a 3-layer medallion architecture (staging → 3NF → star schema), normalising raw Olist e-commerce CSVs into **8 relational tables** with FK constraints enforced at the database level.
- Designed a **5-page Power BI dashboard** tracking **\$13.17M in revenue** across 95K orders, identifying checkout as the highest drop-off stage (**11.56%**), organic search as the top-performing channel (**62x more efficient** than paid search), and VIP customers spending **2.3x more** than first-time buyers; supported by 6 production-grade SQL queries using window functions, recursive CTEs, and NTILE-based RFM segmentation.
- Engineered an **A/B testing framework** with deterministic variant assignment via MD5 hashing in Python and validated a statistically significant **31% conversion lift** (11.71% → 15.36%, $z = 9.24 > 1.96$) through a **SQL-based z-test**, while implementing **CI/CD via GitHub Actions** with automated SQL linting and pytest on every push.

Financial Shock Effects on Hospital Operations | *Python, XGBoost, Feature Engineering*

Jan 2026 – Feb 2026

- Engineered **15+ features** from HCRIS cost reports, including Medicaid share, uncompensated care burden, occupancy rate, staffing intensity, and trained an **XGBoost** model predicting hospital operating margins.
- Stress-tested hospitals under **5–15% Medicaid reimbursement shocks** and compared outcomes across structurally distinct hospital profiles, revealing **private urban (profit/non-profit) hospitals** (Medicaid share <20%, uncompensated care < 5%) held **positive margins** while **urban teaching hospitals** (Medicaid share ~ 40 - 60%, uncompensated care > 10%) became more **financially fragile**.

- Scored and ranked hospitals by resilience ($\Delta\text{margin} = \text{post} - \text{pre shock}$), enabling policymakers to pinpoint the **20% most shock-absorptive** hospital profiles from 2011–2022 margin trend data.

Masked Conditioning for Cascaded Diffusion | *PyTorch, U-Net, Diffusion Models*

Sep 2025 – Dec 2025

- Proposed and implemented a channel-masked conditioning variant of the **cascaded diffusion framework** ($64 \times 64 \rightarrow 128 \times 128$), selectively corrupting **RGB** channels with mask probability 0.35 to improve robustness in **super-resolution** image generation.
- Engineered a two-stage class-conditional diffusion pipeline with **U-Net backbones**, cosine noise schedule over $T=1000$ timesteps, EMA model averaging (decay=0.995), and **DDIM sampling** (200 steps, $\eta=0$), achieving FID ~ 42 at 100 inference steps on a 30-class ImageNet subset ($\sim 33K$ images).
- Concluded both baseline and masked conditioning converge to comparable fidelity, suggesting the channel-masking structure preserves conditioning quality without degrading the augmentation benefit.

Harmful Brain Activity Classification | *ResNet, EfficientNet, TensorFlow*

Feb 2025 – May 2025

- Engineered an **8-channel multi-modal EEG pipeline** fusing raw waveforms with Kaggle spectrograms across **17,089 patient recordings**, reducing input noise via region-aware mel-spectrogram averaging across 4 brain-chain montages (LL, RL, LP, RP).
- Designed clinically-informed spectrogram representations by mapping **19 electrode-pair signals to 4 brain regions**, applying log-scaling and z-score standardization to surface low-frequency structure critical for seizure detection in ICU-monitored patients.
- Benchmarked ResNet vs. **EfficientNetB2** across 3 spectrogram variants using **5-fold GroupKFold cross-validation**, achieving a best KL divergence of **0.43** with combined inputs—a **49% improvement** over single-source baselines—demonstrating effective detection of 6 harmful brain activity patterns.

Auto Insurance Claim Prediction | *R, XGBoost, Tweedie Regression, Cross-Validation*

Oct 2024 – Dec 2024

- Led **R-based modeling and feature selection** for a 60,000-record insurance claim-cost prediction project, handling a highly **semi-continuous dataset** with only **6.8% non-zero claims** and positive costs from \$200 to $\sim$ \$60K.
- Compared Logit + Gamma, Tweedie regression, and XGBoost models using an 80/20 train-test split, 5-fold **cross-validation**, over-dispersion checks, Hosmer-Lemeshow testing, and **AIC-based variable selection**.
- Improved final normalized **Gini index to 0.45** and presented **stakeholder-ready risk drivers**, including exposure, vehicle value, max power, term length, vehicle age, and engineered ratio features for **claim-cost planning**.

Large-Scale Twitter Discourse Analysis: Russia-Ukraine Conflict | *NLP, K-Means*

Jan 2022 – Apr 2022

- Engineered an end-to-end Twitter NLP pipeline that scraped **2.9M tweets** over **15 days** and filtered to **943K English posts**, cutting downstream processing noise by **$\sim 67\%$** .
- Reduced high-dimensional text feature space across **5 NLP representations** (BoW, n-gram, TF-IDF variants) using **PCA**, retaining **90% variance** while shrinking feature complexity for K-Means clustering.
- Identified **3 dominant discourse themes** in $\sim 943K$ war-related tweets using **silhouette- and elbow-optimized K-Means**, surfacing public sentiment patterns across geopolitical, humanitarian, and nuclear threat narratives.

Education

University of Minnesota

Sep 2023 – Aug 2026

Master of Statistics

Minneapolis, United States

Coursework: Data Mining, Computer Vision, Regression Analysis, Advanced Probability, Bayesian Analysis

Indian Institute of Technology Kanpur

Sep 2020 – Jun 2022

Master of Statistics

Kanpur, India

Coursework: Statistical Simulation, Time Series Analysis, Design of Experiment, Applied Stochastic Process

Presidency University

July 2017 – July 2020

Bachelor of Statistics

Kolkata, India

Achievements

- Won HeatMap Hackathon 2026** (BData Inc. $\times$ American Burn Association $\times$ MN Healthcare), built a geospatial systems solution translating burn injury data into an equitable care access analysis, selected over competing teams for real-world strategic impact.
- Received **Academic Excellence Award** for outstanding performance in MSc Statistics, IIT Kanpur.